

Profile

Quantitative biologist bridging computational and wet-lab research, with a track record of building novel molecular workflows, from sequencing to genome editing across diverse biological systems. Skilled at troubleshooting complex systems under real experimental pressure, with a drive to bring this rigor to next-generation diagnostics.

Experience

- 2025-present **Image Analysis Expert** (Data handling and computational analysis), ISTA
- Developed and resolved failures in complex, ML-integrated analysis pipelines, evaluating and integrating new computational tools to support high-throughput quantitative data analysis
 - Trained 500+ researchers on experiment design and analysis pipelines across disciplines
 - Coordinated a zero-disruption, cybersecurity-compliant upgrade of 60+ microscopy workstations aligning imaging facility, IT, and cybersecurity teams for institute-wide infrastructural development
- 2018-2025 **Graduate Research in Biology** (thesis on developmental biophysics), ISTA
- Conceived and led a cross-disciplinary study among 3 institutions and 4 labs – designing experiments, analyzing data and producing a first-author publication in *Nature Communications* ([Naik et al., 2026](#))
 - Performed NGS-informed CRISPR-based perturbations and qPCR /& Western blot validations to dissect cross-system interactions in developing embryos – a systematic approach to causal discovery
 - Co-authored a competitive successful €50K FWF research grant with PI
- 2012-2017 **Undergraduate Research in Biology; CNRS, France and IISER Pune, India**
- Led a 3-lab multidisciplinary study on *Wnt*-mediated tissue homeostasis producing a first-author peer-reviewed publication in *The Company of Biologists' - JEB* ([Naik et al., 2020](#))
 - Studied drug-induced systemic disruption via electrophysiology in primary human and mouse cell cultures for Master's thesis at CNRS, Nice, France

Education

- 2018-2025 **PhD in Biology** (thesis on developmental biophysics), ISTA
- supervised by CP Heisenberg together with E Hannezo at ISTA
- 2012-2017 **Integrated BS-MS (GPA: 8.7), Indian Institute of Science Education and Research Pune, India**
- mentored by Laurent Counillon at CNRS, France & Sanjeev Galande with Shivprasad Patil at IISER Pune

Leadership

- 2019-2024 **Team Lead – PhD Balance Social Media Team**
- Led a 25k-follower global NGO social media team for 5 years overseeing content strategy, coordination, and community engagement while growing audience by 4,000+ followers annually across 500+ posts amplifying discourse on mental health in academia
- 2019-2023 **Organizer – Young Science Seminar and Connecting Intermediate Filaments seminar**
- Co-founded and sustained two international seminar series with a volunteer team over ~100+ sessions – delivering biweekly events,
 - managing international speaker recruitment alongside logistics with 1,000+ participants

Technical Skills

Wet-lab: Electrophysiology, CRISPR-Cas, Cell culture, Advanced Microscopy, NGS analysis, Physical probing
Data & Computing: Python, Unix/HPC cluster computing
AI & Imaging: Stable diffusion, LLM tools, image analysis pipelines
Web & Design: HTML/CSS, WordPress, Canva, Adobe Illustrator

Awards

- 2024 **Award winner & invited speaker**, Gordon Research Conference on IFs, Spain
- 2023 **Invited speaker** at European Intermediate Filaments Conference 2023, Netherlands
- 2020 **FWF DK NanoCell Fellowship**
- 2019-2022 International paper presentations: EMBO, Cambridge CPB, MicroPhysiological Models
- 2018 **Horizon 2020 Marie Skłodowska-Curie Actions Fellowship** for graduate research at ISTA
- 2016 **CNRS Research Scholarship**
- 2013 **KVPY fellowship** by the Department of Science and Technology, Government of India
- 2012 **INSPIRE fellowship** by the Department of Science and Technology, Government of India for undergraduate studies at IISER Pune, India